

Rishi Raj

Doctoral Researcher in Theoretical Physics

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Doctoral researcher in theoretical physics with expertise in high-performance computing, large-scale simulations, and modern ML/AI. Skilled in C++ and Python, I pioneered neural-network applications in high-energy physics, combining mathematical rigor and software-engineering discipline to extract insight from complex data and build efficient predictive models.

PROFESSIONAL EXPERIENCE

Doctoral Researcher

Oct 2023 – Present

Laboratory of Theoretical and High Energy Physics (LPTHE) — Paris, France

- Designed and optimized high-performance symbolic and numerical computation frameworks in Mathematica and Python for large-scale data analysis and model validation — released open-source on [GitHub](#)
- Built and benchmarked custom algorithms integrating combinatorial and graph-theoretic methods, achieving exact agreement between analytical predictions and numerical simulations.
- Engineered scalable pipelines for processing and visualizing large structured datasets, improving computational efficiency and reproducibility in complex modeling tasks.
- Applied mathematical optimization and statistical analysis to quantify convergence, accuracy, and stability across multi-parameter simulation regimes.
- Authored two [publications](#) and presented results at international research forums, demonstrating advanced problem-solving, programming, and data-analysis capabilities.

Teaching Assistant

Jul – Nov 2022

Department of Physics, IIT Madras — Chennai, India

- Supported PH5060 (Computational Physics Lab) by preparing worksheets, grading assignments, and leading tutorials on simulation techniques in Mathematica and Python.
- Supervised PH1030 (Physics Lab I) practical sessions, graded reports, and conducted final examinations for first-year undergraduates.
- Delivered a four-part *Relativity from Symmetries* lecture series for the Horizon Club, simplifying abstract group-theory concepts for early-stage students.

Web Development Freelancer

Jan – Jun 2022

Zenoholics Pvt. Ltd. — Bangalore, India

- Modernized full-stack web applications using React.js, Next.js, and MongoDB, achieving significant performance and scalability gains through architectural refactoring.
- Built CI/CD pipelines with GitHub Actions, Docker, and AWS, deploying containerized microservices with automated testing and monitoring.
- Diagnosed and resolved performance bottlenecks across front- and back-end systems, improving latency and throughput under production workloads.

INTERNSHIPS

Charpak Summer Training Fellow

May – Aug 2023

Sorbonne Université — Paris, France

- Selected for the highly competitive Charpak Lab Fellowship, awarded by the French government to top Indian students for advanced research training in France, hosted jointly by LPTHE and LPENS (ENS Physics Department).
- Explored algorithmic partitioning of multi-center black-hole configuration space using tree-based attractor-flow methods, developing computational routines to analyze phase-space structure and scaling behaviour.

National Initiative on Undergraduate Science (NIUS) Fellow

Oct 2022 – April 2023

Homi Bhabha Centre For Science Education (HBCSE), TIFR — Mumbai, India

- Selected as a NIUS (National Initiative on Undergraduate Science) Physics Fellow at HBCSETIFR, a highly competitive research program nurturing India's top undergraduate researchers.
- Designed and conducted an experiment on magnetically coupled oscillators using low-cost materials and smart-phone based sensing, combining analytical modeling with numerical simulation for data validation, which was published in [The Physics Teacher](#)

EDUCATION

Doctor of Philosophy, PhD in Mathematical Physics

Sorbonne Université — Paris, France

Thesis: *BPS Black Holes and Donaldson-Thomas invariants*.

Oct 2023 – Present

Bachelor and Master of Science, BS & MS in Theoretical Physics

Indian Institute of Technology, IIT Madras — Chennai, India

Graduated with highest distinctions and Minor in Mathematics.

Master's thesis: *Black Holes through the lens of Matrix Theory*.

9.18/10 (Overall), 9.59/10 (Masters) GPA

2018 – 2023

ACADEMIC PROJECTS

Learning Holographic Horizons

Published in *Phys. Rev. D* 

- Applied deep neural networks to model horizon dynamics in holographic systems, linking boundary observables to emergent geometric features.
- Built large-scale numerical-relativity simulations (AdS_4) with pseudospectral solvers and adaptive time-stepping to generate high-fidelity datasets for ML analysis.

Jan – Dec 2023

Black Holes through the lens of Matrix theory

Master's thesis — unpublished, available on [GitHub](#) 

- Developed high-performance C++ simulations of large matrix systems, implementing scalable numerical solvers and benchmarking pipelines to ensure stability and accuracy over long evolutions.
- Analyzed spectral and statistical properties of simulated data using Mathematica, uncovering signatures of chaotic dynamics and random-matrix behavior in complex many-body systems.

Oct 2022 – April 2023

AWARDS AND OTHER ACTIVITIES

4EU+ Ph.D. Summer School on Machine Learning in Physics

@ Niels Bohr Institute, University of Copenhagen

Completed an intensive program on advanced ML methods (BDTs, Bayesian NNs, Transformers, Anomaly Detection) with hands-on model training and evaluation on large datasets.

Jun 2025

1st ROOT Hackathon

@ CERN — Geneva, Switzerland

Contributed bug fixes and code improvements to CERN's ROOT framework during its inaugural open-source hackathon, enhancing tool robustness and maintainability.

Feb 2024

63rd and 64th Institute Day prizes

IIT Madras — Chennai, India

Awarded for outstanding academic and research performance among the graduating batch at IIT Madras.

Aug 2023

Music Club, IIT Madras

IIT Madras — Chennai, India

Organized and coordinated performances for Saarang, the annual cultural festival, developing strong teamwork and event management skills.

Jul 2019 – Nov 2019

Microsoft Codefundo++ Runner Up

Microsoft India – IIT Madras, Chennai, India

Secured top position in a national hackathon for developing innovative solutions in disaster management.

2018

KVPY Fellowship

Indian Institute of Science (IISc) – Bangalore, India

Recipient of the national Kishore Vaigyanik Protsahan Yojana fellowship for aptitude in research in basic sciences.

2017

TECHNICAL SKILLS

Programming Languages: Proficient in Python (scientific and data science applications), Modern C/C++, JavaScript / TypeScript (full-stack systems)

ML & AI Frameworks: PyTorch, sklearn, Hugging Face, Transformers, Pandas visualization with Seaborn, R, ROOT

Scientific Computing: Mathematica, GNU Scientific Library (GSL), Boost C++, Julia, SageMath

Dev Tools & Version Control: Advanced Git workflows, Linux System Administration, VS Code

Full Stack Web Development: Node.js, Flask, FastAPI, React based frameworks like Next.js, MongoDB, SQL

Languages: English (native), French – A1, German – A1 (cont.)